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Deep eutectic solvent (TMAH·5H_2O/Urea) with low viscosity for cellulose dissolution at room temperature

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ABSTRACT

A deep eutectic solvent (DES) formulated with tetramethylammonium hydroxide pentahydrate /urea (TMAH·5H_2O/Urea) was designed for the first time to dissolve cellulose at room temperature. The optimized system, characterized by a 1:3 M ratio, demonstrates the capability to dissolve approximately 7.5 wt% cellulose, boasting a high degree of polymerization (DP = 526). Notably, both the pure DES and 4.0 wt% cellulose/TMAH·5H_2O/Urea mixtures manifests low viscosity, establishing its potential as an effective spinning aid in fiber manufacturing. The structural analyses shows that the cellulose crystal type shifts from type I to type II form, accompanied by a reduction in both crystallinity and DP. A pivotal aspect of this research involves determining Kamlet-Taft parameters for TMAH·5H_2O/Urea-DES with different molar ratios. The results reveal these solvate DESs exhibit the high hydrogen bond basicity, which enables them to easily form hydrogen bonds with hydroxyl groups of cellulose and demonstrate good cellulose solubility. In conclusion, this solvent system presents notable advantages, including straightforward synthesis procedures, low viscosity, and well cellulose solubility, paving the way for new approaches and techniques in cellulose utilization.

1. Introduction

Cellulose, a prevalent renewable resource on Earth, harbors immense potential to emerge as a sustainable alternative to fossil fuels in the production of valuable biomaterials. Known for its extensive hydrogen-bonded networks and highly ordered crystalline structures, cellulose is insoluble in water and conventional organic solvents (Kuang et al., 2008; Liu & Zhang, 2009), posing challenges for its efficient utilization. To date, the quest for an efficient cellulose solvent remains a key focus in cellulose research to unlock its potential for high-value product conversion.

Various solvent systems have been employed in industrial cellulose production, including the viscose and Lyocell processes. The traditional viscose fiber production, employing the NaOH/CS_2 system, offers high controllability but raises concerns due to the emission of harmful substances like CS_2 (B et al., 2014; Klemm et al., 2005; Wang et al., 2012). In the Lyocell process, which utilizes *N*-methylmorpholine-N-oxide (NMMO), offers a more environmentally friendly approach but demands significant energy input for solvent recovery (Rosenau et al., 2001; Zhao et al., 2007). Ionic liquids (ILs) used in the Ion-cell method possess the advantage designability (Swatloski et al., 2002). Tailoring ionic liquid solvents based on the cellulose raw material source can enhance cellulose dissolution capabilities while maintaining environmental sustainability. However, this approach is often cost-prohibitive and presents challenges in solvent recovery and disposal (Fumino et al., 2011; H. Zhang et al., 2005). Given these challenges, there is an urgent need to explore and develop environmentally friendly, cost-effective, and energy-efficient cellulose solvents.

Deep eutectic solvents (DESs) emerge as promising substitutes for ILs, offering similar properties while being more cost-effective and straightforward to synthesize (Moura et al., 2017; Tang et al., 2017; Zdanowicz et al., 2018). The systems consist of a hydrogen bond acceptor (HBA, e.g., quaternary ammonium salts), paired with a hydrogen bond donor (HBD, e.g., carboxylic acids, amines, amides, and polyols). This unique characteristic endows DESs with designable properties akin to those of ILs, allowing for the tailoring of solvents specifically for cellulose dissolution. Previous literatures have reported the suitability and capability of DESs for cellulose solubility, including choline chloride-urea (ChCl-Urea), proline-malic acid, and choline hydroxide/urea (ChOH/Urea) (Francisco et al., 2012; Sharma et al., 2013; Zhong et al., 2022). However, when replicating cellulose dissolution experiments based on existing literature, it was discovered that

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achieving consistent high solubility of cellulose in DESs remains a challenge. By summarizing various cellulose dissolution systems, we have identified certain recurring features. On one hand, successful systems often include urea and alkaline substances, such as NaOH/Urea and ChOH/Urea. On the other hand, these systems are also characterized by the presence of quaternary ammonium salt bases, like NMMO, ChOH/Urea, alkylammonium hydroxides, and imidazolium-ILs. Building on these findings, we hypothesized that a DES incorporating these features could effectively dissolve cellulose. As a result, we opted for tetramethylammonium hydroxide (TMAH·5H_2O) as the HBA, coupled with urea serving as the HBD for our DES. TMAH·5H_2O has a diverse range of applications in semiconductors, optics, and chemistry, including photographic developer and polarographic analysis. Urea, a natural biometabolite commonly used as a fertilizer, is readily available and cost-effective. These factors collectively render TMAH·5H_2O and urea attractive choices for our cellulose dissolution study.

The objective of this study was to design a DES by mixing the two components, and to investigate its effectiveness in dissolving cellulose. We initially discussed the effects of DES molar ratio and dissolution temperature on cellulose solubility. The Kamlet–Taft solvatochromic parameters were investigated to evaluate the polarity of DESs in different molar ratios in the present study. Viscosity measurements of the DES and cellulose solutions were conducted to assess their potential for producing regenerated cellulose materials. Furthermore, we characterized the regenerated cellulose (RC) using Fourier transform infrared (FTIR) spectroscopy, X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and thermogravimetric analysis (TGA). the RC was characterized by the means of Fourier transform infrared (FTIR) spectroscopy, X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and thermogravimetric analysis (TGA). The present study proposes a promising and cost-effective solvent for cellulose, offering new insights for efficient cellulose solubilization.

2. Experimental materials and methods

2.1. Materials

Bleached softwood kraft pulp (BSKP) used in this study was supplied by Northwood Pulp and Timber Ltd. in Canada. Prior to use, the pulp was oven-dried overnight at 60 °C under vacuum. Urea (99 %), Tetramethylammonium hydroxide pentahydrate (TMAH·5H_2O) (97 %) methanol (≥99.9 %), and microcrystalline cellulose (MCC) were purchased from Shanghai Macklin Biochemical Co. Ltd. The dyes, Reichardt's dye (RD) (90 %), 4-nitroaniline (NA) (99 %), and N, N-diethyl-4-nitroaniline (DENA) (>98 %), were purchased from Sigma-Aldrich, Germany. Cupriethylenediamine (CED) solution (>99 %) was obtained from National Pulp and Paper Research Institute., Ltd.

2.2. Synthesis of DES

TMAH·5H_2O and Urea were mixed in various molar ratios (1:1, 1:1.25, 1:1.5, 1:1.75, 1:2, 1:3, 1:3.25, and 1:3.5) and stirred at 150 rpm using a magnetic stirrer at room temperature until a transparent and homogeneous liquid solution was achieved (Fig. S1).

2.3. Solubility of cellulose and regeneration of cellulose

BSKP was added into the DES and stirred at 150 rpm using a magnetic stirrer at specified temperatures (10, 20, 25, 30, 40, and 50 °C). The solubility of BSKP in the DES was assessed using a polarized optical microscope (POM, Olympus EP50). Samples were collected and observed within 30 min after each addition of BSKP. The dark field view of the POM confirmed the complete dissolution of cellulose. The described procedures were repeated until additional BSKP could no longer be dissolved, even after a period exceeding 2 h. Solubility was calculated based on the total quantity of BSKP successfully dissolved within the DES. Each experiment was performed in triplicate, and the results were averaged across these repetitions.

RC was prepared by adding deionized water to the cellulose solution. The resulting mixture was then filtered, and the collected RC underwent multiple rinses with deionized water to ensure the complete removal of any residual DES. Finally, the RC was vacuum-dried at 60 °C overnight before undergoing further characterization.

2.4. Kamlet-Taft parameters measurement of DESs

The Kamlet-Taft parameters were determined from the absorption peaks of the three dyes, Reichardt's dye (RD, Mw = 551.69 g/mol), N, N-diethyl-4-nitro-aniline (DENA, Mw = 194.23 g/mol), and 4-nitroaniline (NA, Mw = 138.12 g/mol). In detail, the dyes were respectively dissolved in methanol and stored in amber glass vials. Each time 5 mL dye solution was added into a 5 mL centrifuge tube. The methanol was evaporated by vacuum drying at 40 °C for 2 days. Then, 3 mL DES was put into the dried dye, and stirred to dissolve the dye completely. and was transferred into a 1 cm^2 quartz cuvette. The absorption spectrum of the mixture was checked using UV–visible spectrophotometer in the wavelength range of 300–700 nm. The maximum absorbance of the UV–vis spectra (UV 2310 II) was controlled in the range of 0.4–2.0 by adjusting the volume of the dye solution added in the vials. The experimental temperature was set at 10 °C, 25 °C, and 50 °C. The calculation of hydrogen bond acidity (α), hydrogen bond basicity (β), and dipolarity-polarizability (π^*) values use the following Eqs. (1)–(5) (Hauru et al., 2012):

$$\upsilon_{max} = \frac{1}{\lambda_{max} \times 10^{-4}} \tag{1}$$

$$E_T(30) = \frac{28592}{\lambda_{max}(RD)} \tag{2}$$

$$\pi^* = 0.314(27.52 - \upsilon_{max}(DENA)) \tag{3}$$

$$\alpha = 0.0649E_T(30) - 2.03 - 0.72\pi^* \tag{4}$$

$$\beta = \frac{(1.035\upsilon_{max}(DENA) + 2.64 - \upsilon_{max}(NA))}{2.8} \tag{5}$$

2.5. Measurements

FTIR spectra were collected using a Fourier transform infrared (FTIR) spectroscopy (NICOLET iS10). Each spectrum was recorded in the range of 4000–650 cm^{-1} with a resolution of 4 cm^{-1}, accumulating 16 scans. All samples were prepared with KBr pellets, and the pure KBr spectrum served as the background spectrum for baseline correction.

X-ray diffraction (XRD) measurements were conducted using a diffractometer (Bruker D8 ADVANCE, Bruker, America) equipped with Cu-Kα radiation source (λ = 1.54 Å). The operating conditions were set at 40 kV and 40 mA, with a scanning rate of 5°/min within the 2θ range of 5°- 45°. All spectra have been subjected to baseline correction and normalization. The crystallinity index of the samples was calculated using the following equation: (Segal et al., 1959)

$$CrI(\%) = \frac{Ac}{Ac + Aa} \times 100 \tag{6}$$

where CrI is the crystallinity index of the RC, Ac is the area for crystalline regions, and Aa is the area for amorphous regions.

The degree of polymerization (DP) of both BSKP and RC was determined using an Ubbelohde viscometer (Ma et al., 2018). The procedure involved dissolving a measured mass of dried BSKP and RC samples in a 0.5 M CED solution. The outflow time of the samples was measured with the viscometer at 25 °C, and the characteristic viscosity [η] was then calculated. Each experiment was repeated in triplicate, and the final

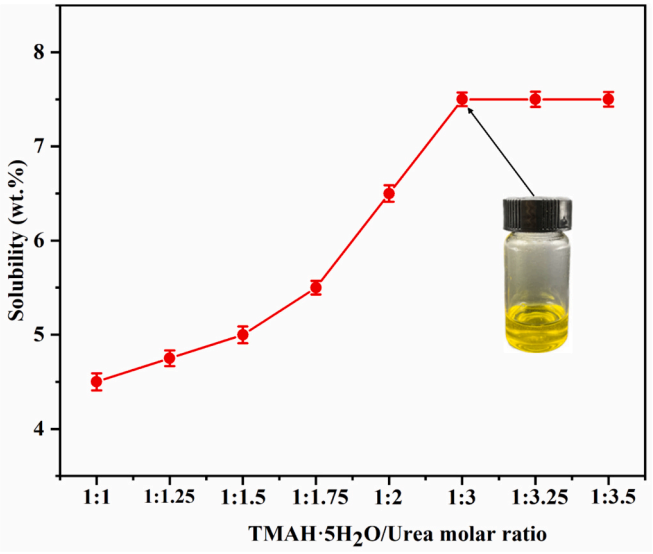


Fig. 1. Solubility of cellulose in the TMAH·5H_2O/Urea solvent with various molar ratios. The insert photo shows the homogeneous cellulose/TMAH·5H_2O/Urea solution.

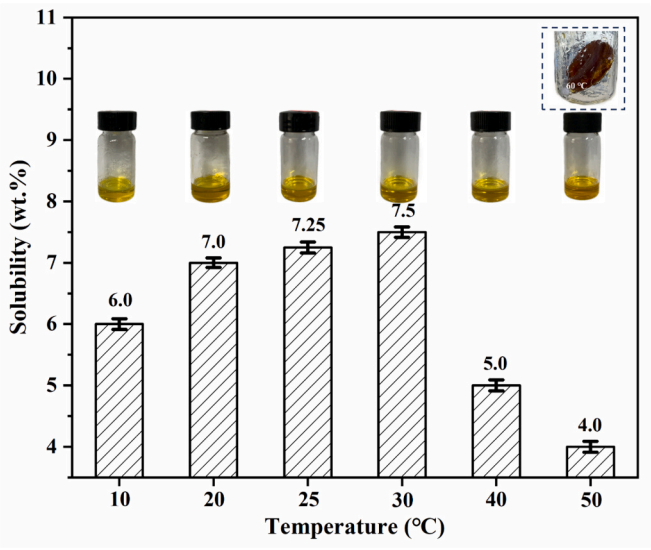


Fig. 2. Solubility of BSKP in TMAH·5H_2O/Urea (1:3) aqueous system as a function of temperature. The cellulose solutions obtained at different temperatures are displayed at the top of figure accordingly. The inset figures in the upper right corner shows the inability of this DES to dissolve cellulose when the temperature is further increased to 60 °C.

result was derived as the average of these measurements. The relationship between the characteristic viscosity $[\eta]$ of cellulose and the average DP satisfied the following equation (Marx & Physics, 1955):

$$DP^{0.905} = 0.75[\eta] \quad (7)$$

The rheological properties of cellulose solutions were investigated using a rotational rheometer (Anton Paar MCR302, Austria) equipped with a parallel plate device (25 mm plate diameter, 1 mm gap size). Viscosity measurements were conducted across a temperature range spanning 30 to 80 °C, with a heating rate of 5 °C/ min and a consistent shear rate of 4 s^{-1}. The steady shear experiment covered a shear rate range of 0.01–100 s^{-1}, while the dynamic oscillatory shear experiment encompassed a frequency range of 0.1–100 rad/s. Before each frequency sweep in the oscillatory shear measurements, a strain sweep ranging from 0.1 to 30 % at a constant frequency of 1 rad/s was performed to identify the linear viscoelastic regime. A 5 % strain was selected within the frequency range of 0.1–100 rad/s to ensure that the measurements remained within the linear viscoelastic region.

X-ray photoelectron spectra (XPS) were collected on a Thermo Scientific K-Alpha XPS system (Thermo Fisher Scientific, UK) equipped with a micro-focused, monochromatic Al Kα X-ray source (1486.6 eV). Curve fitting calculations were carried out using XPSPEAK software version 4.1.

Thermal properties of DESs, BSKP and RC were investigated using thermogravimetric analysis (TGA) (TGASDTA851e, Mettler Toledo, Switzerland). The samples were heated from 30 to 800 °C at a heating rate of 10 °C/min under a nitrogen atmosphere.

3. Results and discussion

3.1. Solubility of cellulose in TMAH·5H_2O/Urea

DES was prepared by mixing HBA and HBD in appropriate molar ratios to form a transparent and homogeneous solution. Previous studies have demonstrated that a homogeneous liquid could be achieved within the TMAH·5H_2O/Urea molar ratio range of 1:1 to 1:3.5. Therefore, TMAH·5H_2O/Urea molar ratios of 1:1, 1:1.25, 1:1.5, 1:1.75, 1:2, 1:3, 1:3.25, and 1:3.5 were selected for analyzing cellulose solubility. Fig. 1 illustrates how cellulose solubility is dependent on the TMAH·5H_2O/Urea molar ratio at room temperature. It was shown that the solubility of cellulose was notably high (≥4.5 wt%) across all investigated molar ratios. As the proportion of urea increased, an increased solubility of cellulose in the DES was also observed. The maximum solubility of cellulose (~7.5 wt%) was achieved at a TMAH·5H_2O/Urea molar ratio of 1:3. The insert photo in Fig. 1 confirms the formation of a homogeneous cellulose/TMAH·5H_2O/Urea solution. In-situ POM observations further demonstrated complete dissolution of the cellulose in the solvent within 1 h (Fig. S2). Within the system, TMAH·5H_2O and urea interact via strong hydrogen bonding to form DES. It is hypothesized that there is a strong interaction between the -NH_2 group of urea and the tetramethylammonium hydroxide anion. Initially, this interaction predominates at lower urea concentrations. As urea concentration increases, this interaction persists as a fundamental component, alongside other interactions such as -NH_2···C=O, which collectively maximize hydrogen bonding within the system.

Here, TMAH·5H_2O was chosen as the alkylammonium hydroxide for DES synthesis. This choice is based on the commercial availability of TMAH, which is typically supplied in the form of an aqueous solution or with crystalline water content. Water, as a constant component in TMAH·5H_2O, was left untreated during the experimental process. It was implied that the water content in the system varied with different TMAH·5H_2O/Urea molar ratios. Specifically, an increase in the urea proportion resulted in a decrease in the water content of the DES. The solubility of BSKP in the TMAH·5H_2O/Urea aqueous system increased as the water content decreased. This could be explained by the fact that water molecules participate in the formation of hydrogen bonds within the system (Rumyantsev et al., 2018). The interaction of water molecules and solvent system competes with that of cellulose molecules and solvent system, leading to a reduction in the solubility of cellulose. Removing water may potentially enhance the solubility of TMAH·5H_2O/Urea-DES for cellulose. Theoretical investigations have indicated that the presence of water speeds up hydrogen-bond dynamics and reduces heterogeneity in DES (Srinivasan et al., 2021). The influence of water on the solubility of cellulose in TMAH·5H_2O/Urea-DES is less pronounced compared to that in ILs, where cellulose becomes insoluble with water content exceeding 1.0 wt% (Swatloski et al., 2002). The TMAH·5H_2O/Urea-DES exhibited favorable cellulose solubility even in the presence of 20–40 wt% water content. It means that the energy consumption for water removal during the preparation and recycling procedures of the solvent system can be reduced.

Table 1
Kamlet-Taft parameters of DES composed of TMAH·5H_2O (HBA) and Urea (HBD) for different molar ratios at various temperatures.

Solvent	Molar ratio	T/ °C	E_T (30)	π^*	α	β	β-α
TMAH·5H_2O/ Urea	1:1	25	51.33	1.10	0.50	1.02	0.52
	1:2	25	51.38	1.12	0.49	1.14	0.65
	1:3	25	51.42	1.13	0.48	1.36	0.88
	1:3	10	51.56	1.10	0.52	1.19	0.68
	1:3	50	51.89	1.11	0.53	0.92	0.40

Fig. 2 illustrates the relationship between temperature and the solubility of BSKP in the TMAH·5H_2O/Urea system at a molar ratio of 1:3, a ratio where higher cellulose solubility is observed (as shown in Fig. 1). The solubility of cellulose in this system exhibited a clear temperature dependency. As the temperature increased initially, the solubility of cellulose in the TMAH·5H_2O/Urea system also increased. Maximum solubility was achieved at room temperature with approximately 7.5 wt % BSKP could be dissolved. When the temperature raised beyond room temperature, the solubility of cellulose began to decrease. Optimal solubility for cellulose was demonstrated within the temperature range of 25–30 °C. Higher temperatures cause significant degradation of cellulose, resulting in the breaking of molecular chains and changes in its chemical structure. Notably, the color of the cellulose solution darkened at 50 °C, and at 60 °C, cellulose hardly dissolves, yielding a dark brown solid (as shown in the inset photo in Fig. 2). Additionally, the solubility of microcrystalline cellulose (MCC) with a low DP was also investigated in the TMAH·5H_2O/Urea. At room temperature, approximately 12 wt% of MCC could be completely dissolved within 1 h (Fig. S2). Compared to traditional NMMO and ILs, the TMAH·5H_2O/Urea provides milder conditions for cellulose dissolution. In ILs, temperatures for cellulose dissolution typically range from 80 to 110 °C (Isik, Sardon, Mecerreyes, 2014a),(Isik, Sardon, Mecerreyes, 2014b), while for NMMO monohydrate, they are above 80 °C (Rosenau & French, 2021). Moreover, the TMAH·5H_2O/Urea system demonstrates a greater solubility than the low-temperature NaOH/Urea aqueous system, which only offers limited solubility for cellulose with a relatively low DP. The promising solubility of the TMAH·5H_2O/Urea system for high molecular weight cellulose is crucial for advancing processes aimed at producing regenerated cellulose materials with specific desired properties.

3.2. Kamlet–Taft parameters

The Kamlet-Taft polarity scale, crucial for assessing the polarity of DESs, quantifies the α, β and π^* of DES solvents. The β value generally evaluates the hydrogen bond acceptor ability, which exhibits a strong correlation with cellulose solubility (Xu et al., 2010). Additionally, the cellulose dissolving ability of DESs is linked to the $\beta - \alpha$ parameter (net basicity). Specifically, DESs with β parameter higher than 0.80 and net basicity in the range of 0.35–0.90 are capable to dissolve cellulose. Solvents like ILs with $\beta > 0.8$ easily form hydrogen bonds with the hydroxyl groups of cellulose, resulting in favorable cellulose solubility (Brandt et al., 2013).

Table 1 summarizes the Kamlet-Taft parameters of the DESs investigated in this study, all of which meet the desired criteria. The β parameter values increased with the increasing urea molar ratio, indicating enhanced hydrogen bond basicity. This phenomenon correlated with the excellent cellulose solubility observed in TMAH·5H_2O/Urea-DES, consistent with our experimental findings. Previous reports have indicated that urea enhances cellulose solubility in alkali (Cai et al., 2008; Jiang et al., 2014) and TBAH aqueous solvent systems (Perkins et al., 2013; Sun et al., 2013; Zhang et al., 2019) by stabilizing isolated cellulose chains in the solution. In the case of the ChCl/Urea (1:2) system, vibrational analysis and molecular dynamic simulations have highlighted a strong interaction between the chlorine anion and the -NH_2 group of urea. It is reasonable to infer a similar interaction between the -NH_2 group of urea and the anion of tetramethylammonium hydroxide in the TMAH·5H_2O/Urea-DES system, leading to the formation of moderate hydrogen bonds. Hence, urea's role in the TMAH·5H_2O/Urea system differs from its role in alkali and TBAH systems. In this system, urea acts as the HBD in conjunction with TMAH·5H_2O as the HBA to form a DES through strong hydrogen bonding interactions. The dissolution of cellulose in TMAH·5H_2O/Urea-DES is postulated to proceed via an electron acceptance and supply mechanism involving the hydrogen and oxygen atoms of cellulose hydroxyl groups. The anion (OH-) of TMAH·5H_2O/Urea-DES induces an electron-induced interaction with the hydrogen atom of the electron-deficient group of the cellulose hydroxyl group. This disrupts the original intermolecular hydrogen bonding in the cellulose structure. Simultaneously, the cation of the quaternary ammonium salt reacts with the oxygen atom of the cellulose hydroxyl group to form a new hydrogen bond between the solvent and cellulose (as shown in Fig. 3).

The α value slightly decreased with the increase of urea proportion, possibly due to reduced water content in the system. Water, known for its low hydrogen bond basicity ($\beta = 0.18$) and high hydrogen bond acidity ($\alpha = 1.17$) (Hauru, Hummel, King, Kilpelinen, & Sixta, 2012; Kamlet, Abboud, Abraham, & Taft, 1983), forms strong interactions with the OH- ions of TMAH·5H_2O/Urea, affecting the hydrogen bond basicity of the solvent. Excessive water molecules could compete with cellulose in the formation of hydrogen bonds, which led to reducing the solubility of cellulose in TMAH·5H_2O/Urea-DES. Additionally, the influence of temperature on the K-T parameter was investigated. It was found that the β value decreased at temperature above 50 °C, which may be related to the thermal decomposition of the active ingredient in DES.

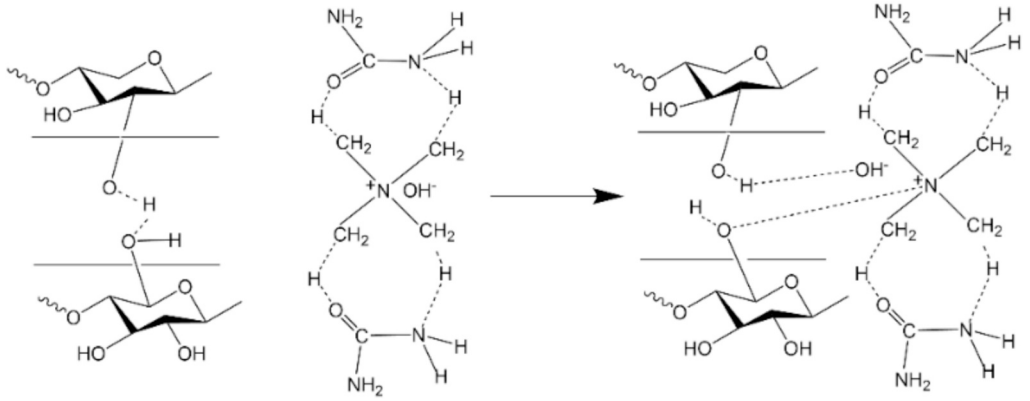


Fig. 3. Possible H-bond interactions mechanism between cellulose and TMAH·5H_2O/Urea-DES.

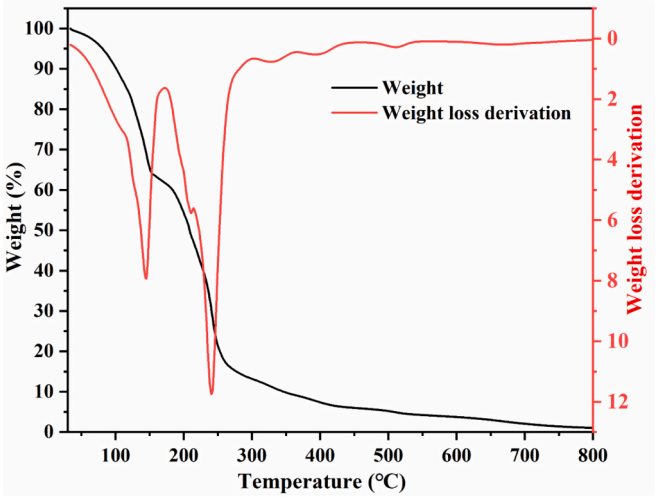


Fig. 4. Thermogravimetry (TGA) and differential thermogravimetry (DTG) curves of TMAH·5H_2O/Urea-DES in a nitrogen atmosphere.

3.3. Thermal stability of TMAH·5H_2O/Urea

As illustrated by the TGA curve and DTG curve in Fig. 4, the thermal characteristics of the TMAH·5H_2O/Urea-DES were presented. An initial weight loss of about 10 % below 100 °C was attributed to water evaporation in the system. The system exhibited excellent thermal stability below 100 °C, with the thermal decomposition occurring above this temperature. Therefore, to ensure optimal conditions for cellulose dissolution, storage of cellulose solutions, and preparation of regenerated cellulose materials, it was advisable to maintain temperature below 100 °C. According to Fig. 2, showing cellulose solubility in the TMAH·5H_2O/Urea system at different temperatures, the ideal range for cellulose dissolution falls approximately between 25 and 30 °C, aligning with the room temperature range. This temperature range is well-suited for subsequent processes involved in the preparation of regenerated cellulose materials.

3.4. Rheological characteristics analysis of cellulose/TMAH·5H_2O/Urea solutions

The rheological properties of cellulose solutions are crucial in setting processing parameter for RC materials preparation (Elsayed et al., 2021). Both steady and complex viscosity measurements were conducted on pure TMAH·5H_2O/Urea-DES solvent and 4.0 wt% cellulose solutions at various temperatures. Fig. 5 (a) illustrates the apparent viscosity (η) of solutions in relation to temperature. TMAH·5H_2O/Urea-DES demonstrated relatively low η values across the tested temperature range. At room temperature (30 °C), the η value for the 4.0 wt% cellulose solution notably increased to 0.248 Pa·s. This value is lower than those reported for [AMIM][Cl] (0.790 Pa·s at 30 °C) (Fendt et al., 2011) and [DBUH][$CH_3CH_2OCH_2COO$] (0.812 Pa·s at 30 °C) (X. Wang et al., 2023). As the temperature increased, the η values of the cellulose solutions decreased significantly. This is attributed to the enhanced thermal motion of molecules at higher temperatures, leading to a reduction in intermolecular interactions within the solution.

Fig. 5 (b) displays the steady shear solution viscosity as a function of shear rate. Based on the viscosity curve, several parameters such as non-Newtonian index (n), zero shear viscosity (η_0), and structure index (Δ_η) can be calculated. The zero shear viscosity (η_0) represents viscosity as shear rates tend toward zero and can be calculated using the Carreau-Yasuda I model (Bird et al., 1987):

$$\frac{\eta-\eta_\infty}{\eta_0-\eta_\infty}=\frac{1}{[1+(\lambda\cdot\dot{\gamma})^{\alpha}]^{\frac{1-n}{\alpha}}} \tag{8}$$

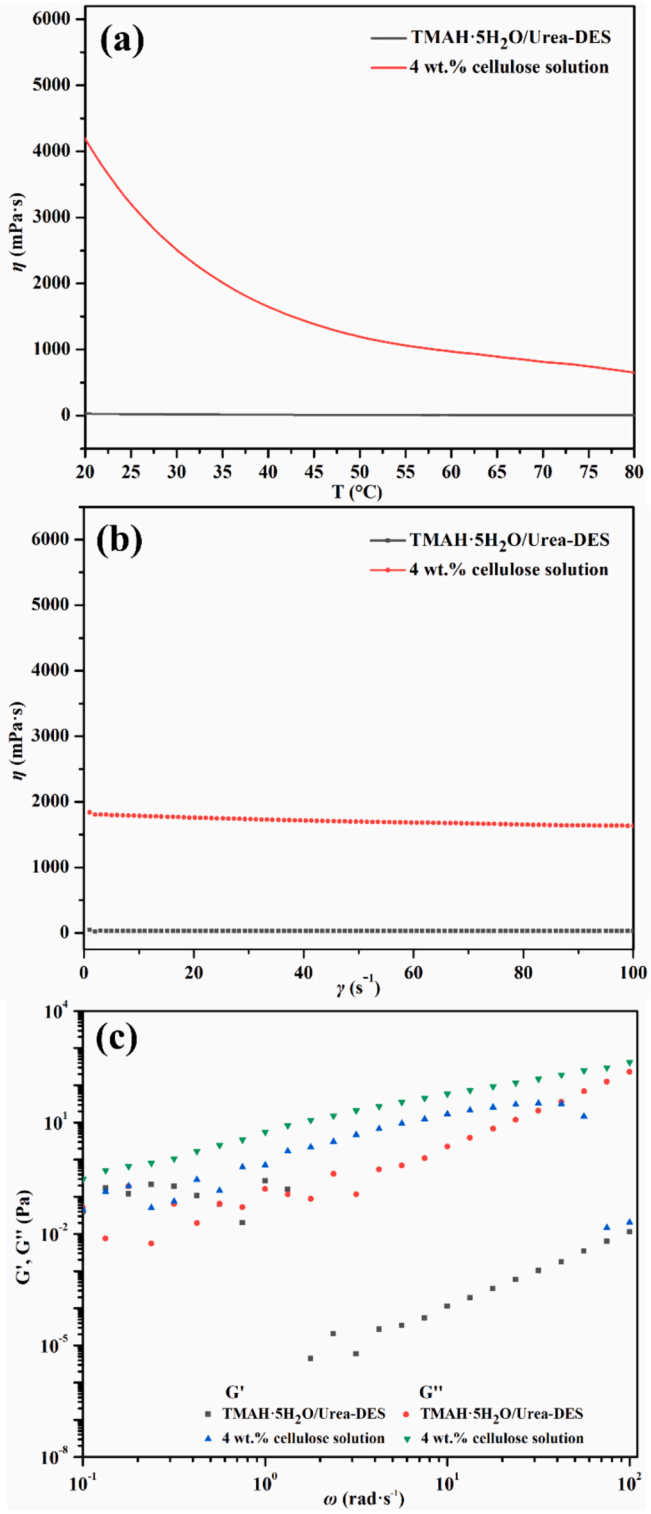


Fig. 5. (a) Steady shear viscosity (η) as a function of (a) temperature and (b) $\dot{\gamma}$ for pure TMAH·5H_2O/Urea-DES solvent and BSKP solution with the cellulose concentration of 4.0 wt%; (c) storage modulus (G'), loss modulus (G'') curves for pure TMAH·5H_2O/Urea-DES solvent and BSKP solution with the cellulose concentration of 4.0 wt%.

where η_0 represents the zero-shear viscosity, η_∞ is the viscosity at an infinite shear rate, and $\dot{\gamma}$ is the shear strain.

At an infinitely high shear rate, the polymer undergoes decomposition, resulting in a calculated viscosity of zero.

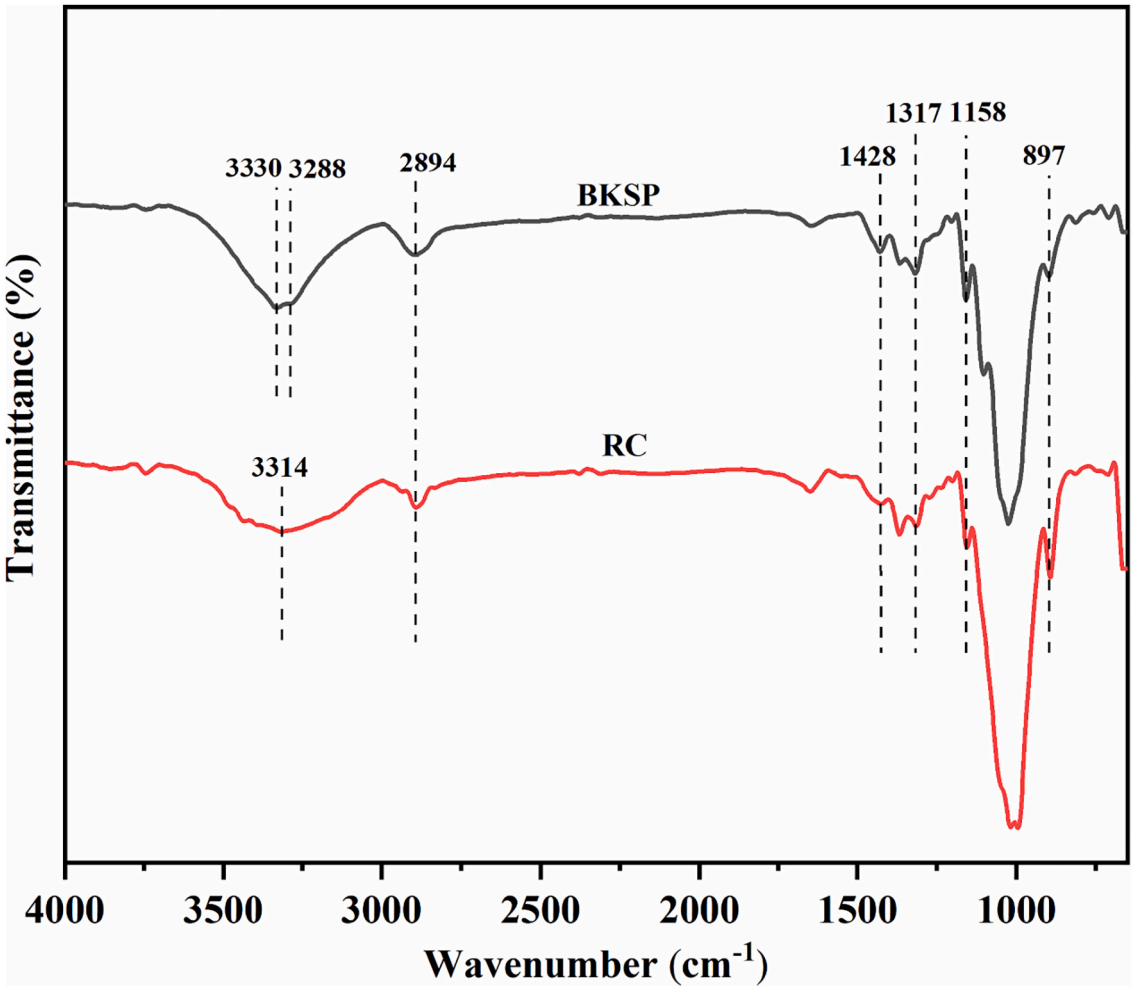


Fig. 6. FTIR spectra of Bleached softwood kraft pulp (BSKP) and cellulose sample (RC) regenerated from TMAH·5H_2O/Urea-DES solution after dissolution at room temperature.

$$\eta = \frac{\eta_0}{\left[1 + (\lambda \cdot \dot{\gamma})^{\alpha}\right]^{\frac{1-n}{\alpha}}} \tag{9}$$

By fitting the viscosity curve with the Carreau-Yasuda I model and plotting viscosity against the shear rate, we can determine the value of the zero-shear viscosity (η_0). The results demonstrated that the pure DES solvent exhibited Newtonian behavior, with a linear relationship between shear stress and shear rate. In contrast, the cellulose/TMAH·5H_2O/Urea solution displayed typical shear-thinning behavior, with decreasing viscosity at increased shear rate. This behavior is commonly observed in cellulose and ionic liquids solutions (Chen et al., 2011; Lu et al., 2013). The η_0 of the 4.0 wt% cellulose solution was significantly higher, measuring at 1854.1 mPa·s. It indicated an enhancement of intermolecular interactions between cellulose and DES. The cellulose solution experienced shear thinning at higher shear rates, causing disintegration of association or aggregation structures of cellulose chains (Yue et al., 2021). As the shear rate increased, cellulose chains gradually untangled and aligned in the direction of shear, resulting in a change in the nature of the solution from a Newtonian fluid to a non-Newtonian fluid. It was worth noting that despite exhibiting shear-thinning behavior, the cellulose solution remained transparent, with no discernible undissolved particles visible under an optical microscope (Fig. S2) (Lee et al., 2019). Similar phenomenon have been observed in other polymer solutions, such as Lyocell solutions and conventional highly-viscosity polymer solutions (Elsayed et al., 2021; Lu et al., 2013).

The concept of the structural viscosity index (Δ_η) is introduced to assess the degree of structuring within cellulose solutions. This index is computed as follows:

$$\Delta_\eta = \left(-\frac{d \lg_\eta}{d \dot{\gamma}^{\frac{1}{2}}} \right) \times 100 \tag{10}$$

where Δ_η is determined by the linear fit of lgη versus $\dot{\gamma}^{1/2}$. For the 4.0 wt % cellulose solution, the values of $\Delta_{\eta 1}$(0.58) and $\Delta_{\eta 2}$(0.62) represented the degree of structuring in the shear rate ranges of 1–40 s^{-1} and 40–100 s^{-1}, respectively. In the non-Newtonian region, a positive value of the structural index represents a degree of structuring in the shear-thinning fluid. A lower Δ_η value indicates a less structured fluid, which is favorable for spinnability (Appaw et al., 2007). Generally, the Δ_η value is crucial for processing and shaping cellulose materials, especially in spinning processes. Lowering Δ_η values can help mitigate the tendency of the spinning solution to form structures and improve its spinnability (Cai et al., 2013). Compared to previously reported cellulose solvents dissolved in [DBUH][$CH_3CH_2OCH_2COO$] (X. Wang et al., 2023), the 4.0 wt% cellulose solution examined in this study exhibited significantly diminished Δ_η values, suggesting that the solution could serve as a valuable aid in facilitating fiber production during the spinning process.

The relationship between the storage modulus (G') and loss modulus (G'') in the context of angular frequency (ω) was examined for both the

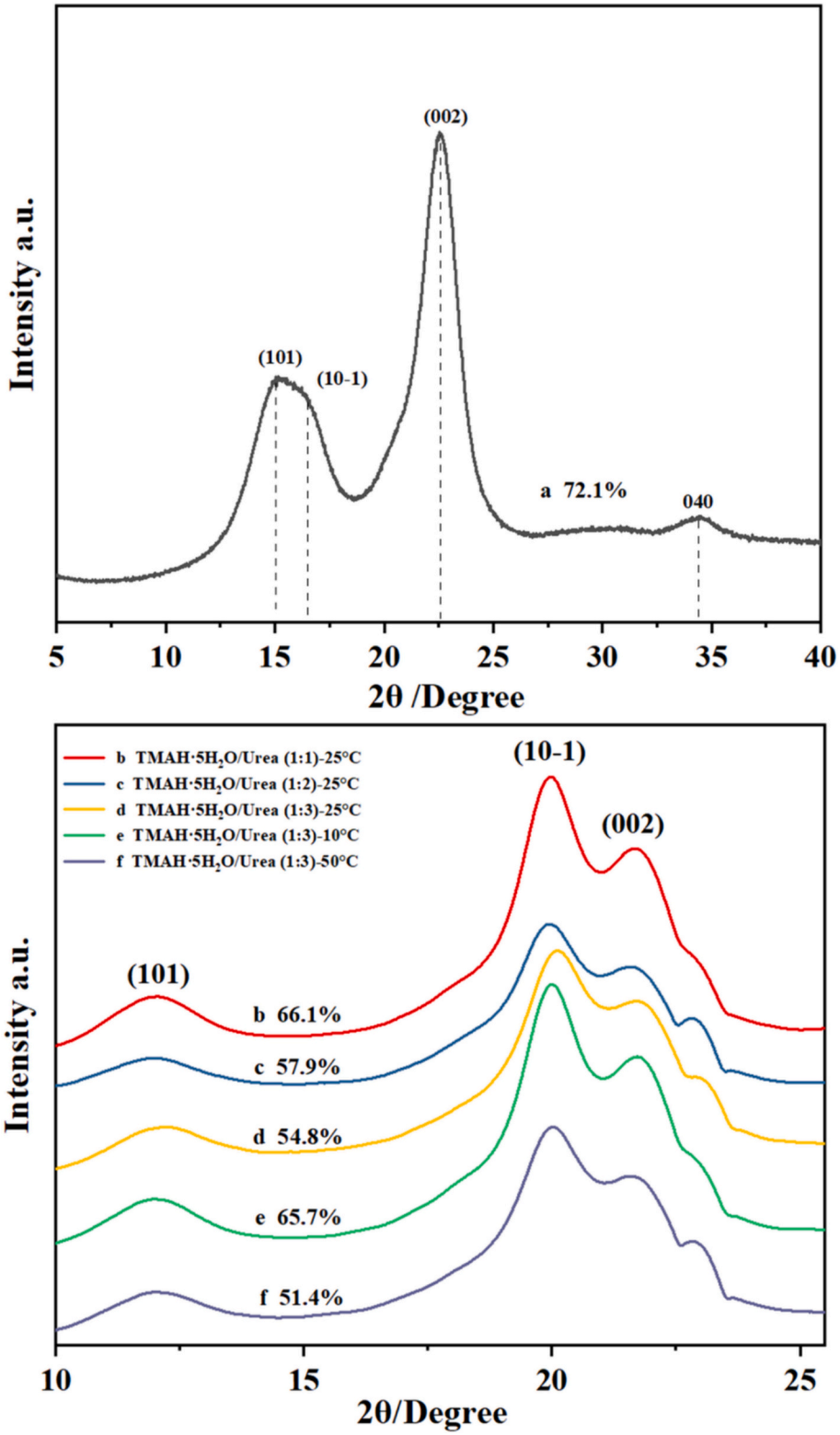


Fig. 7. XRD patterns and crystallinity of BSKP (a) and the cellulose regenerated from (b) TMAH·5H_2O/Urea (molar ratio 1:1) at 25 °C, (c) TMAH·5H_2O/Urea (molar ratio 1:2) at 25 °C, (d) TMAH·5H_2O/Urea (molar ratio 1:3) at 25 °C, (e) TMAH·5H_2O/Urea (molar ratio 1:3) at 10 °C, and (f) TMAH·5H_2O/Urea (molar ratio 1:3) at 50 °C.

pure DES solvent and a 4.0 wt% cellulose solution, as depicted in Fig. 5 (c). The 4.0 wt% cellulose solution exhibited more pronounced elastic behavior than the pure solvent at identical vibration frequences. At this cellulose concentration, G' was consistently smaller than G'' over all measured angular frequencies, suggesting characteristics of a viscous-like liquid. Throughout this process, the solutions deformed gradually, with most energy dissipation occurring via viscous flow.

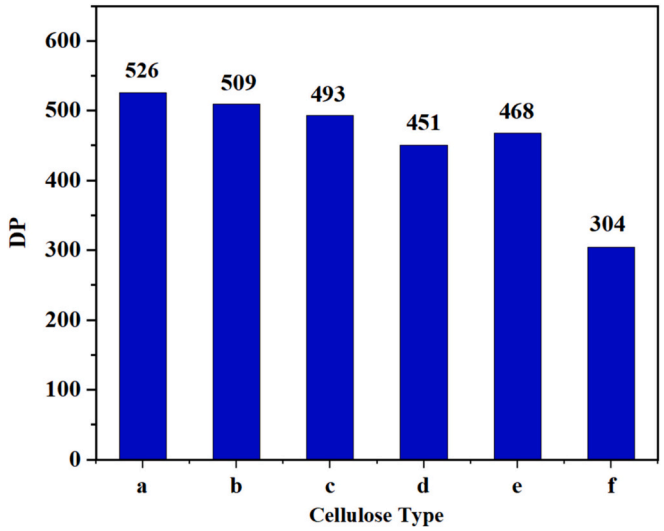


Fig. 8. DP for BSKP (a) and the cellulose regenerated from (b) TMAH·5H_2O/Urea (molar ratio 1:1) at 25 °C, (c) TMAH·5H_2O/Urea (molar ratio 1:2) at 25 °C, (d) TMAH·5H_2O/Urea (molar ratio 1:3) at 25 °C, (e) TMAH·5H_2O/Urea (molar ratio 1:3) at 10 °C, and (f) TMAH·5H_2O/Urea (molar ratio 1:3) at 50 °C.

3.5. Structure of regenerated cellulose from TMAH·5H_2O/Urea

Fig. 6 illustrates the typical FTIR spectra of both BSKP and RC samples obtained from the TMAH·5H_2O/Urea system within the wavelength range of 4000–650 cm^{-1}. The characteristic peaks of BSKP at 3650–3200, 2894, 1428–1317, 1158, and 897 cm^{-1} correspond to O—H vibration, symmetric and asymmetric vibration of CH_2 groups, CH_2 and CH scissoring vibration, C-O-C stretching vibration, and β-1, 4 glycosidic linkages of cellulose, respectively (Lateef et al., 2009; Li et al., 2019; Ruan et al., 2004; Sun et al., 2004). Notably, the spectral patterns of RC bore a striking resemblance to those of BSKP, implying that no chemical reactions occurred during the processes of cellulose dissolution and subsequent regeneration. Upon closer examination, a shoulder peak at 3288 cm^{-1} was identified in BSKP but was absent in RC. The absorption peaks observed at 3330 cm^{-1} and 3288 cm^{-1} in BSKP merged into a broad absorption peak at 3314 cm^{-1} in RC. This broad peak could still be attributed to the O—H vibrations within cellulose, suggesting that the disruption of hydrogen bonds had occurred during the dissolution and regeneration processes (Zhou et al., 2001).

XRD analysis was employed to elucidate changes in the aggregate structure and crystallinity of cellulose (Fig. 7). The XRD pattern of BSKP exhibited characteristic diffraction peaks at approximately 2θ = 15°, 16.4°, 22.5°, and 34.4°, corresponding to the lattice planes (101), (10−1), (002), and (040), respectively (Peng et al., 2013). These peaks signified the presence of the cellulose I crystalline phase in BSKP. In contrast, the XRD pattern of RC obtained from cellulose/ TMAH·5H_2O/Urea solution displayed diffraction peaks at 2θ values of 12.1°, 20.5°, and 21.6°, which were associated with the (101), (10–1), and (002) crystallographic planes of cellulose II (Cao et al., 2009). This observation suggested a transformation from cellulose I to cellulose II during the dissolution and regeneration processes, likely due to the facile association or aggregation of cellulose chains. Additionally, the XRD pattern of RC showed a decrease in peak intensity, and the corresponding crystallinity ranged from 51.4 to 66.1 %, lower than that of BSKP (72.1 %). These results pointed to the rapid disruption of the original crystalline structure of cellulose by TMAH·5H_2O/Urea, leading to the cleavage of both inter-molecular and intra-molecular hydrogen bonds during the dissolution process.

Table 2
Characteristic parameters obtained from DTG profiles of BSKP and RC samples during the combustion and pyrolysis processes.

Samples	Onset temperature (°C)	End temperature (°C)	Residual mass (%)	Weight change (%)	Peak temperature of DTG (°C)
BSKP	323.5	378.8	12.6	85.53	359.4
RC	315.1	359.8	33.3	64.96	338.0

The DP of the RC could be affected by various factors including solvent systems, dissolution temperature, and dissolution time. Fig. 8 illustrated that RC exhibited lower DP values compared to BSKP, indicating a higher degree of cellulose degradation during the dissolution process. This finding is consistent with previous literature reports (A. Xu et al., 2010). The basic nature of the TMAH·5H_2O/Urea system is known to induce a stripping reaction of the reducing end groups on the cellulose molecular chain, ultimately leading to cellulose degradation (Ishiguri et al., 2005). Also, high temperature was responsible for decreased polymerization degree. This degradation phenomenon can have implications for the performance of RC materials.

3.6. Thermal stability of BSKP and RC

The TGA and DTG curves presented in Fig. 9, along with the characteristic parameters shown in Table 2, provide insights into the thermal stability of both BSKP and RC. The cellulose degradation process can be categorized into three stages: water evaporation, decomposition, and carbonization. The initial weight loss at lower temperatures primarily resulted from water evaporation within the cellulose, which was influenced by the initial moisture content. The second significant weight loss observed for both BSKP and RC occurred above 315 °C, corresponding to cellulose decomposition. This gradual weight loss process continued as the temperature increased to 360 °C. Table 2 clearly indicated that the second phase of weight loss for BSKP occurred within the temperature range of 323.5 to 378.8 °C, while for RC, it was observed between 315.1

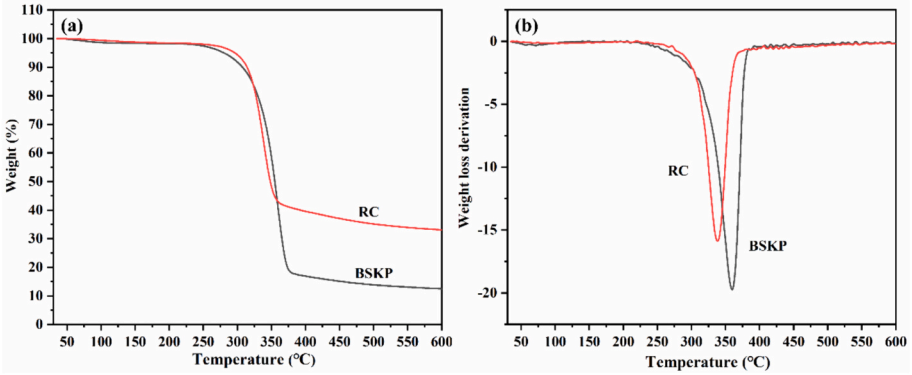


Fig. 9. (a) TGA and (b) DTG curves of raw cellulose (BSKP) and regenerated cellulose (RC).



Fig. 10. Scanning electron microscope (SEM) photos of BSKP (a, b) and RC (c, d).
Scale bar: 500 μm (a); 10.0 μm (b)-(d).

and 359.8 °C. During the decomposition stage, RC displayed a slightly lower onset temperature of 315.1 °C compared to BSKP, which had an onset temperature of 323.5 °C. The peak temperature in the DTG curve represented the maximum thermal degradation temperature. BSKP exhibited a maximum thermal degradation temperature of 359.4 °C, while RC had a lower maximum temperature of 338.0 °C. This indicated that RC had lower thermal stability, likely due to the disruption of hydrogen bonds during the dissolution and regeneration process. These differences in thermal decomposition between the two cellulose samples are consistent with their disparities in crystallinity (Kargarzadeh et al., 2012). Specifically, the reduction in crystallinity of RC correspond to its lower thermal stability. Fig. 9 (a) also revealed that RC exhibited a notably higher residual mass (non-volatile carbonaceous material) than BSKP after thermal decomposition. A similar phenomenon was reported by Xu et al (Xu et al., 2019). The diminished thermal stability of RC could potentially enhance low-activation energy reactions, in which the cellulose dehydration and levoglucosan generation could promote char formation (Velásquez-Cock et al., 2016; Xu et al., 2019). The higher char yield of RC can also be related to the lower crystallinity of cellulose. The amorphous region of RC is more prone to form char residue during pyrolysis. SEM analyses of BSKP and RC exhibited distinct differences (Fig. 10). The BSKP had a dense and well-organized structure (Fig. 10 (a), (b)), whereas the surfaces of the regenerated cellulose samples appeared disordered and rough, lacking the fibrous-shaped cellulose structure (Fig. 10 (c), (d)).

3.7. XPS analysis of the BSKP and RC

Chemical surface characterization of the BSKP and RC was conducted using X-ray photoelectron spectroscopy (XPS), a powerful analytical technique for determining the elemental composition of a sample's surface. By analyzing high-resolution spectra, XPS provides valuable insights into the distribution of elements on the surface. The XPS wide-scan spectra of the cellulose samples are depicted in Fig. 11 (a) and 11 (d). The results revealed that the surface of both BSKP and RC primarily consists of carbon and oxygen, with no detected hydrogen. The measured oxygen-to-carbon (O/C) ratio for BSKP was 0.6707, while for RC, it is 0.6616. It suggested that the cellulose dissolution and subsequent regeneration processes induced changes in the surface composition, resulting in a diminished oxygen content.

The analysis of binding energies and chemical shifts of the C1s peaks provides valuable insights into their surrounding chemical environment. In the high-resolution spectra of the total C1s peaks in BSKP, three distinct component peaks are observed at approximately 283.1 eV, 284.9 eV, and 286.5 eV. These peaks correspond to C1 (carbon bonded exclusively to carbon and/or hydrogen atoms, C-C/C-H), C2 (carbon bonded to a single oxygen atom other than a carbonyl oxygen, C-OH), and C3 (carbon bonded to two non-carbonyl oxygen atoms or a single carbonyl oxygen atom, O-C-O/C=O), respectively (Matuana et al., 2001). The XPS C1s spectra of BSKP and RC are shown in Fig. 11 (b) and 11 (e), with corresponding data provided in Table 3. The results indicated that the chemical bond types of the macromolecules on the cellulose surface remained unchanged after the dissolution and regeneration process. However, there were notable variations in the ratio of each chemical bond. In the C1s spectrum of BSKP, a distinct peak for C═O, a relatively strong peak for C—O, and a negligible peak for C-C/C-H bonds were observed. Conversely, RC exhibited a significant increase in the intensity of the C-C/C-H peak. This suggested that the

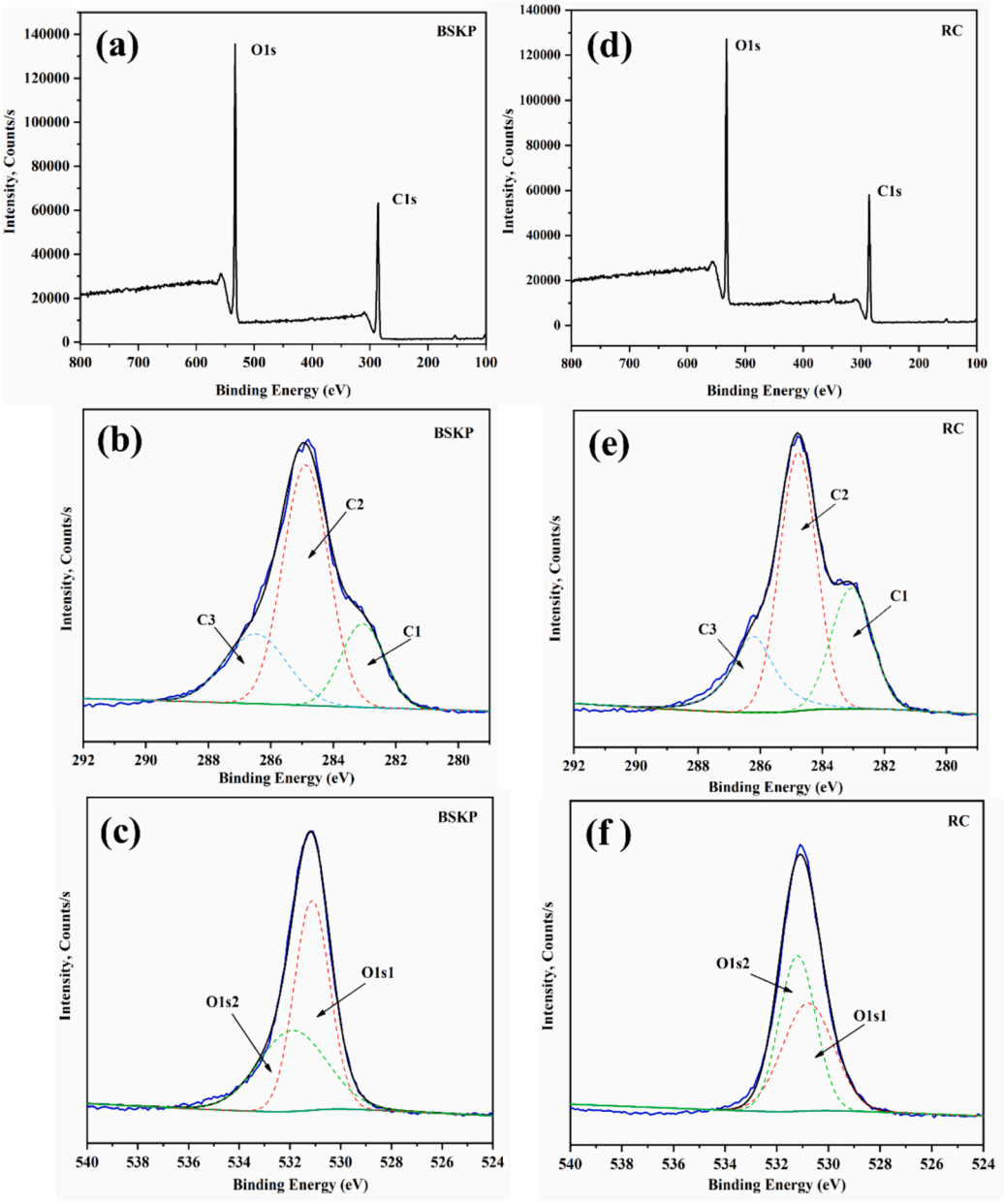


Fig. 11. XPS survey spectra of BSKP (a) and RC (d). XPS C1s spectra of (b) BSKP and (e) RC (where C1 is C-C/C-H, C2 is C-OH and C3 is O-C-O/C=O). XPS O1s spectra of (c) BSKP and (f) RC (where O1s1 is C-OH···O and O1s2 is C-OH).

Table 3
Peak assignments and peak area ratios of the three kinds of C1s peaks used in curve fitting for BSKP and RC.

Samples	Peak location /eV			Peak area ratio /%		
	C1	C2	C3	C1	C2	C3
BSKP	283.1	284.9	286.5	19.06	57.9	23.0
RC	283.1	284.8	286.2	27.00	52.5	20.5

Table 4
Peak assignments and peak area ratios of the two kinds of O1s peaks used in curve fitting for BSKP and RC.

Samples	Peak location /eV		Peak area ratio /%	
	O1s1	O1s2	O1s1	O1s2
BSKP	531.1	531.9	56.9	43.1
RC	530.8	531.2	49.3	50.8

dissolution and regeneration process can affect the relative abundance of C-C/C-H bonds on the cellulose surface. Furthermore, the bonding energies of bonds closely related to hydrogen bonding, such as C-OH and O-C-O, demonstrated a tendency to decrease. This phenomenon may be attributed to the presence of hydroxyl groups in the DES, which could disrupt the bonding of these specific bonds on the cellulose macromolecule. Consequently, the overall bonding energy of the cellulose macromolecule decreased as a result of these alterations.

The XPS O1s spectra of BSKP and RC are depicted in Fig. 11 (c) and 11 (f), with detailed information provided in Table 4. Oxygen atoms within cellulose molecules are primarily linked to carbon atoms through C—O bonds. Certain oxygen atoms participate in the formation of hydrogen bonds, leading to variations in the binding energy of chemical bonds compared to free hydroxyl groups that are not involved in

hydrogen bonding interactions. The O1s spectrum peak can be deconvoluted into two peaks: O1s1 (C-OH···O) and O1s2 (C-OH) (Hou et al., 2023). The C—O bond in C-OH···O exhibits a lower binding energy due to the attraction of oxygen negative ions. The relative contents of O1s1 and O1s2 peaks are calculated based on their respective peak areas. The intricate hydrogen-bonded network structure formed by numerous hydroxyl groups in the cellulose macromolecule directly contributes to the formation of a dense and highly crystalline cellulose structure. The higher crystallinity of BSKP corresponded to the relatively higher content of its O1s1 peak. Conversely, the decrease in crystallinity of RC aligned with the reduction in the relative content of its O1s1 peak. Accordingly, the relative content of the O1s1 peak may be indicative of the relative content of hydroxyl oxygen atoms involved in hydrogen bonding, reflecting the relative content of the crystalline region in cellulose. On the other hand, the relative content of the O1s2 peak reflected the relative content of free hydroxyl oxygen atoms in cellulose, representing the relative content of the amorphous region in cellulose. The decrease in crystallinity was underscored by the elevation in the relative content of free hydroxyl groups, leading to an increase in its O1s2 peak to 50.8 %, accompanied by a decrease in the relative content of the O1s1 peak to 49.3 %. Overall, the XPS spectra confirm the composition of cellulose and the alterations in its hydrogen bonding network during the dissolution and regeneration process. Further investigation is necessary to fully understand the specific changes in chemical bonds and their relationship to the dissolution and regeneration process of cellulose in the TMAH·5H_2O/Urea-DES system.

4. Conclusions

In this study, we explored a new cellulose solvent based on tetramethylammonium hydroxide pentahydrate /urea (TMAH·5H_2O/Urea) deep eutectic solvent (DES), using TMAH·5H_2O as the HBA and Urea as the HBD. The system's synthesis process was straightforward. It was observed that the TMAH·5H_2O/Urea system with a 1:3 M ratio could dissolve a minimum of ~7.5 wt% cellulose with a DP of 526. Maximum solubility occurred at room temperature within 1 h in laboratory conditions, whereas under heating, the solvents showed reduced cellulose dissolution. Kamlet-Taft parameters of the DESs were measured, and the results indicated that all TMAH·5H_2O/Urea-DESs have a strong hydrogen bond basicity with the β value of higher than 0.80. Rheological results revealed a notable correlation between solution viscosity and cellulose concentration. The viscosity of the 4.0 wt% cellulose solution decreased significantly with rising temperature. The cellulose/TMAH·5H_2O/Urea solution maintained low viscosity values across the tested temperature range compared to the ionic liquids. Further FTIR, XRD, and XPS analysis showed cellulose and regenerated cellulose had similar structures, but cellulose's crystalline form transformed from type I to type II after dissolution. Additionally, there was a reduction in crystallinity and polymerization. A dissolution mechanism for cellulose was proposed based on this. It was suggested that the anion (OH-) of TMAH·5H_2O/Urea-DES induced an electron-induced interaction with the hydrogen atom of the electron-deficient group of the cellulose hydroxyl group, leading to the disruption of the original intermolecular hydrogen bonding in the cellulose structure. Simultaneously, the cation of the quaternary ammonium salt formed a new hydrogen bond with the oxygen atom of the cellulose hydroxyl group, facilitating efficient cellulose dissolution in the process. The comprehensive analysis of the dissolution mechanism provides valuable insights into the interactions between cellulose and the DES components, which can facilitate the development of future strategies for cellulose dissolution.

CRediT authorship contribution statement

Li Li: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Maozhi Zhang:** Writing – review & editing. **Yun Feng:** Writing – review & editing. **Xun Zhang:** Writing – review & editing, Funding acquisition. **Feng Xu:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbpol.2024.122260.

References

Appaw, C., Gilbert, R. D., Khan, S. A., & Kadla, J. F. J. B. (2007). Viscoelastic behavior of cellulose acetate in a mixed solvent system. *Biomacromolecules, 8*(5), 1541–1547. https://doi.org/10.1021/bm0611681

B, Q. M. A., A, L. C., A, L. H., C, C. T. B., B, L. Z., & B, Y. N. (2014). A process for enhancing the accessibility and reactivity of hardwood Kraft-based dissolving pulp for viscose rayon production by cellulase treatment. *Bioresource Technology, 154*(2), 109–113. https://doi.org/10.1016/j.biortech.2013.12.040

Bird, R. B., Armstrong, R. C., & Hassager, O. (1987). Dynamics of polymeric liquids. In *Vol. 1: Fluid mechanics*. https://doi.org/10.1063/1.3057567

Brandt, A., Grasvik, J., Hallett, W., & T. (2013). Deconstruction of lignocellulosic biomass with ionic liquids. *Green Chemistry, 15*(3), 550–583. https://doi.org/10.1039/C2GC36364J

Cai, J., Fei, P., Xiong, Z., Shi, Y., Yan, K., & Xiong, H. J. C.p. (2013). Surface acetylation of bamboo cellulose: Preparation and rheological properties. *Carbohydrate Polymers, 92*(1), 11–18. https://doi.org/10.1016/j.carbpol.2012.09.059

Cai, J., Zhang, L., Liu, S., Liu, Y., Xu, X., Chen, X., & Cheng, H. J. M. (2008). Dynamic self-assembly induced rapid dissolution of cellulose at low temperatures. *Macromolecules, 41*(23), 9345–9351. https://doi.org/10.1021/ma801110g

Cao, Y., Wu, J., Zhang, J., Li, H., Zhang, Y., & He, J. J. C. E. J. (2009). Room temperature ionic liquids (RTILs): A new and versatile platform for cellulose processing and derivatization. *Chemical Engineering Journal, 147*(1), 13–21. https://doi.org/10.1016/j.cej.2008.11.011

Chen, X., Zhang, Y., Wang, H., Wang, S.-W., Liang, S., & Colby, R. H. J. J.o. R. (2011). Solution rheology of cellulose in 1-butyl-3-methyl imidazolium chloride. *Journal of Rheology, 55*(3), 485–494. https://doi.org/10.1122/1.3553032

Elsayed, S., Hummel, M., Sawada, D., Guizani, C., Rissanen, M., & Sixta, H. J. C. (2021). Superbase-based protic ionic liquids for cellulose filament spinning. *Cellulose, 28*, 533–547. https://doi.org/10.1007/s10570-020-03505-y

Fendt, S., Padmanabhan, S., Blanch, H. W., Prausnitz, J. M. J. J.o. C., & Data, E. (2011). Viscosities of acetate or chloride-based ionic liquids and some of their mixtures with water or other common solvents. *Journal of Chemical & Engineering Data, 56*(1), 31–34. https://doi.org/10.1021/je1007235

Francisco, M., Van Den Bruinhorst, A., & Kroon, M. C. J. G.c. (2012). New natural and renewable low transition temperature mixtures (LTTMs): Screening as solvents for lignocellulosic biomass processing. *Green Chemistry, 14*(8), 2153–2157. https://doi.org/10.1039/c2gc35660k

Fumino, K., Peppel, T., Geppert-Rybczyńska, M., Zaitsau, D. H., Lehmann, J. K., Verevkin, S. P., & Ludwig, R. J. P. C. C. P. (2011). The influence of hydrogen bonding on the physical properties of ionic liquids. *Physical Chemistry Chemical Physics, 13*(31), 14064–14075. https://doi.org/10.1039/c1cp20732f

Hauru, L. K. J., Hummel, M., King, A. W. T., Kilpelinen, I., & Sixta, H. (2012). Role of solvent parameters in the regeneration of cellulose from ionic liquid solutions. *Biomacromolecules, 13*(9), 2896–2905. https://doi.org/10.1021/bm300912y

Hou, R., Jin, Z., Sun, D., Shi, B., Wang, L., Shan, X. J. J.o. A., & Compounds. (2023). Carbon nanotubes doped cellulose nanocomposite film for high current flexible piezoelectric nanogenerators. *Journal of Alloys and Compounds, 171422*. https://doi.org/10.1016/j.jallcom.2023.171422

Ishiguri, F., Masubuchi, N., Yokota, S., & Yoshizawa, N. J. J.o. W. S. (2005). Changes in the physical and chemical properties of six Japanese softwoods caused by lengthy

smoke-heating treatment. *Journal of Wood Science, 51*, 161–166. https://doi.org/10.1007/s10086-004-0620-8

Isik, M., Sardon, H., & Mecerreyes, D. J. I.j.o.m.s. (2014b). *Ionic liquids and cellulose: dissolution, chemical modification and preparation of new cellulosic materials. 15(7)* pp. 11922–11940). https://doi.org/10.3390/ijms150711922

Isik, M., Sardon, H., & Mecerreyes, D. J. I.j.o.m.s. (2014a). Ionic liquids and cellulose: Dissolution, chemical modification and preparation of new cellulosic materials. *International Journal of Molecular Sciences, 15*(7), 11922–11940. https://doi.org/10.3390/ijms150711922

Jiang, Z., Fang, Y., Xiang, J., Ma, Y., Lu, A., Kang, H., & Zhang, L. J. T. J.o. P. C. B. (2014). Intermolecular interactions and 3D structure in cellulose–NaOH–urea aqueous system. *The Journal of Physical Chemistry B, 118*(34), 10250–10257. https://doi.org/10.1021/jp501408e

Kamlet, M. J., Abboud, J. L. M., Abraham, M. H., & Taft, R. W. (1983). Linear solvation energy relationships. 23. A comprehensive collection of the solvatochromic parameters,. pi.*,. alpha., and. beta., and some methods for simplifying the generalized solvatochromic equation. *The Journal of Organic Chemistry, 48*(17), 2877–2887. https://doi.org/10.1021/jo00165a018

Kargarzadeh, H., Ahmad, I., Abdullah, I., Dufresne, A., Zainudin, S. Y., & Sheltami, R. M. J. C. (2012). Effects of hydrolysis conditions on the morphology, crystallinity, and thermal stability of cellulose nanocrystals extracted from kenaf bast fibers. *Cellulose, 19*(3), 855–866. https://doi.org/10.1007/s10570-012-9684-6

Klemm, D., Heublein, B., Fink, H. P., & Bohn, A. J. A.c.i.e. (2005). Cellulose: Fascinating biopolymer and sustainable raw material. *Angewandte Chemie International Edition, 44*(22), 3358–3393. https://doi.org/10.1002/anie.200460587

Kuang, Q.-L., Zhao, J.-C., Niu, Y.-H., Zhang, J., & Wang, Z.-G. J. T. J.o. P. C. B. (2008). Celluloses in an ionic liquid: The rheological properties of the solutions spanning the dilute and semidilute regimes. *The Journal of Physical Chemistry B, 112*(33), 10234–10240. https://doi.org/10.1021/jp804167n

Lateef, H., Grimes, S., Kewcharoenwong, P., Feinberg, B. J. J., & o. C. T., & Biotechnology. (2009). Separation and recovery of cellulose and lignin using ionic liquids: A process for recovery from paper-based waste. *Journal of Chemical Technology and Biotechnology, 84*(12), 1818–1827. https://doi.org/10.1002/jctb.2251

Lee, Y. J., Lee, S. J., Jeong, S. W., Kim, H.-c., Oh, T. H., Lee, S. G. J. F., & Polymers. (2019). Structure and mechanical properties of regenerated cellulose fibers wet-spun from ionic liquid/cosolvent systems. Fibers and Polymers, 20, 501–511. doi:https://doi.org/10.1007/s12221-019-8335-z.

Li, J., Liu, X., Zheng, Q., Chen, L., Huang, L., Ni, Y., & Ouyang, X. J. C. (2019). Urea/NaOH system for enhancing the removal of hemicellulose from cellulosic fibers. *Cellulose, 26*, 6393–6400. https://doi.org/10.1007/s10570-019-02587-7

Liu, S., & Zhang, L. J. C. (2009). Effects of polymer concentration and coagulation temperature on the properties of regenerated cellulose films prepared from LiOH/urea solution. *Cellulose, 16*, 189–198. https://doi.org/10.1007/s10570-008-9268-7

Lu, F., Song, J., Cheng, B.-W., Ji, X.-J., & Wang, L.-J. J. C. (2013). Viscoelasticity and rheology in the regimes from dilute to concentrated in cellulose 1-ethyl-3-methylimidazolium acetate solutions. *Cellulose, 20*, 1343–1352. https://doi.org/10.1007/s10570-013-9885-7

Ma, X., Huang, H., Huang, F., Long, Y., Cao, S., Chen, L., & Ni, Y. (2018). Synergistic effects of enzyme pretreatment for hemicellulose separation from paper-grade pulp in ionic liquid/water. *Cellulose, 25*. https://doi.org/10.1007/s10570-018-1856-6

Marx, V. M. J. D. M. C. M. C., & Physics. (1955). Viskosimetrische molekulargewichtsbestimmung von cellulose in kupfer-äethylendiamin. *Macromolecular Chemistry and Physics, 16*(1), 157–176. https://doi.org/10.1002/macp.1955.020160118

Matuana, L., Balatinecz, J., Sodhi, R., Park, C. J. W. S., & Technology. (2001). Surface characterization of esterified cellulosic fibers by XPS and FTIR spectroscopy. *Wood Science and Technology, 35*(3), 191–201. https://doi.org/10.1007/s002260100097

Moura, L., Moufawad, T., Ferreira, M., Bricout, H., Tilloy, S., Monflier, E., & Fourmentin, S. J. E. C. L. (2017). Deep eutectic solvents as green absorbents of volatile organic pollutants. *Environmental Chemistry Letters, 15*, 747–753. https://doi.org/10.1007/s10311-017-0654-y

Peng, Y., Gardner, D. J., Han, Y., Kiziltas, A., Cai, Z., & Tshabalala, M. A. J. C. (2013). Influence of drying method on the material properties of nanocellulose I: Thermostability and crystallinity. *Cellulose, 20*, 2379–2392. https://doi.org/10.1007/s10570-013-0019-z

Perkins, S. L., Painter, P., Colina, C. M. J. T. J., & o. P. C. B.. (2013). Molecular dynamic simulations and vibrational analysis of an ionic liquid analogue. *The Journal of Physical Chemistry B, 117*(35), 10250–10260. https://doi.org/10.1021/jp404619x

Rosenau, T., & French, A. D. J. C. (2021). N-Methylmorpholine-N-oxide (NMMO): Hazards in practice and pitfalls in theory. *Cellulose, 28*(10), 5985–5990. https://doi.org/10.1007/s10570-021-03860-4

Rosenau, T., Potthast, A., Sixta, H., Kosma, P. J. P., & i. p. s.. (2001). The chemistry of side reactions and byproduct formation in the system NMMO/cellulose (Lyocell process). *Progress in Polymer Science, 26*(9), 1763–1837. https://doi.org/10.1016/s0079-6700(01)00023-5

Ruan, D., Zhang, L., Mao, Y., Zeng, M., Li, X. J. J., & o. M. S.. (2004). Microporous membranes prepared from cellulose in NaOH/thiourea aqueous solution. *Journal of Membrane Science, 241*(2), 265–274. https://doi.org/10.1016/j.memsci.2004.05.019

Rumyantsev, M., Rumyantsev, S., Kalagaev, I. Y. J. T. J., & o. P. C. B.. (2018). Effect of water on the activation thermodynamics of deep eutectic solvents based on carboxybetaine and choline. *The Journal of Physical Chemistry B, 122*(22), 5951–5960. https://doi.org/10.1021/acs.jpcb.8b01218

Segal, L., Creely, J. J., Martin, A., Jr., Conrad, C. J. T., & r. j.. (1959). An empirical method for estimating the degree of crystallinity of native cellulose using the X-ray diffractometer. *Textile Research Journal, 29*(10), 786–794. https://doi.org/10.1177/004051755902901003

Sharma, M., Mukesh, C., Mondal, D., & Prasad, K. J. R. A. (2013). Dissolution of α-chitin in deep eutectic solvents. *RSC Advances, 3*(39), 18149–18155. https://doi.org/10.1039/c3ra43404d

Srinivasan, H., Sharma, V., Mitra, S. J. T. J., & o. C. P.. (2021). Water accelerates the hydrogen-bond dynamics and abates heterogeneity in deep eutectic solvent based on acetamide and lithium perchlorate. *The Journal of Chemical Physics, 155*(2). https://doi.org/10.1063/5.0054942

Sun, H., Li, Y., Wu, X., Li, G. J. J., & o. m. m.. (2013). Theoretical study on the structures and properties of mixtures of urea and choline chloride. *Journal of Molecular Modeling, 19*, 2433–2441. https://doi.org/10.1007/s00894-013-1791-2

Sun, X. F., Sun, R., Tomkinson, J., Baird, M. J. P., & d., & stability.. (2004). Degradation of wheat straw lignin and hemicellulosic polymers by a totally chlorine-free method. *Polymer Degradation and Stability, 83*(1), 47–57. https://doi.org/10.1016/s0141-3910(03)00205-2

Swatloski, R. P., Spear, S. K., Holbrey, J. D., Rogers, R. D. J. J., & o. t. A. c. s.. (2002). Dissolution of cellose with ionic liquids. *Journal of the American Chemical Society, 124*(18), 4974–4975. https://doi.org/10.1021/ja025790m

Tang, X., Zuo, M., Li, Z., Liu, H., Xiong, C., Zeng, X., & Lei, T. J. C. (2017). Green processing of lignocellulosic biomass and its derivatives in deep eutectic solvents. *ChemSusChem, 10*(13), 2696–2706. https://doi.org/10.1002/cssc.201700457

Velásquez-Cock, J., Gañán, P., Posada, P., Castro, C., Serpa, A., Putaux, J.-L., & Products.. (2016). Influence of combined mechanical treatments on the morphology and structure of cellulose nanofibrils: Thermal and mechanical properties of the resulting films. *Industrial Crops and Products, 85*, 1–10. https://doi.org/10.1016/j.indcrop.2016.02.036

Wang, H., Gurau, G., & Rogers, R. D. J. C. S. R. (2012). Ionic liquid processing of cellulose. *Chemical Society Reviews, 41*(4), 1519–1537. https://doi.org/10.1039/c2cs15311d

Wang, X., Zheng, W., Guo, Z., Nawaz, H., You, T., Li, X., & Xu, F. J. G. C. (2023). Rheological characteristics of novel cellulose/superbase-derived ionic liquid solutions and the coagulation process towards regenerated cellulose films. *Green Chemistry, 25*(4), 1597–1610. https://doi.org/10.1039/d2gc03278c

Xu, A., Wang, J., & Wang, H. (2010). Effects of anionic structure and lithium salts addition on the dissolution of cellulose in 1-butyl-3-methylimidazolium-based ionic liquid solvent systems. *Green Chemistry*. https://doi.org/10.1039/b916882f

Xu, K., Xiao, Y., Cao, Y., Peng, S., Fan, M., Wang, K. J. C., & p.. (2019). Dissolution of cellulose in 1-allyl-3-methylimidazolium methyl phosphonate ionic liquid and its composite system with Na2PHO3. *Green Chemistry, 209*, 382–388. https://doi.org/10.1039/b916882f

Yue, W., Zhang, L., Deng, L., Guo, Y., Xu, Q., Peng, W., & Liang, S. J. G. C. (2021). Co-dissolution of cellulose and silk fibroin in levulinic acid-derived protic ionic liquids for composited membrane and fiber preparation. *Green Chemistry, 23*(23), 9669–9682. https://doi.org/10.1039/d1gc02837e

Zdanowicz, M., Wilpiszewska, K., Spychaj, T. J. C., & p.. (2018). Deep eutectic solvents for polysaccharides processing. A review. *Carbohydrate Polymers, 200*, 361–380. https://doi.org/10.1016/j.carbpol.2018.07.078

Zhang, H., Wu, J., Zhang, J., & He, J. J. M. (2005). 1-Allyl-3-methylimidazolium chloride room temperature ionic liquid: A new and powerful nonderivatizing solvent for cellulose. *Macromolecules, 38*(20), 8272–8277. https://doi.org/10.1021/ma0505676

Zhang, J., Wang, M., Li, W., Wei, W., Li, J., Jiang, M., & Zhou, Z. J. C. (2019). TBAH/Urea/H 2 O solvent for room temperature wet-spinning of cellulose and optimization of drawing process. *Cellulose (London, England), 26*, 6959–6977. https://doi.org/10.1007/s10570-019-02536-4

Zhao, H., Kwak, J. H., Wang, Y., Franz, J. A., White, J. M., & Holladay, J. E. J. C. P. (2007). Interactions between cellulose and N-methylmorpholine-N-oxide. *Carbohydrate Polymers, 67*(1), 97–103. https://doi.org/10.1016/j.carbpol.2006.04.019

Zhong, Y., Wu, J., Kang, H., & Liu, R. J. G. C. (2022). Choline hydroxide based deep eutectic solvent for dissolving cellulose. *Green Chemistry, 24*(6), 2464–2475. https://doi.org/10.1039/d1gc04130d

Zhou, Tashiro, K., Hongo, T., Shirataki, H., Yamane, C., & Ii, T. J. M. (2001). Influence of water on structure and mechanical properties of regenerated cellulose studied by an organized combination of infrared spectra, X-ray diffraction, and dynamic viscoelastic data measured as functions of temperature and humidity. *Macromolecules, 34*(5), 1274–1280. https://doi.org/10.1021/ma001507x